TECHNICAL REPORT

Global AI Jobs Analysis

An Exploratory Data Analysis of 90,000 AI Job Records Using Python

Item	Detail
Project	Global AI Jobs: compensation, work conditions, hiring and market indicators
Dataset	global_ai_jobs.csv (90,000 records, 35 variables, 2020 to 2026)
Tools	Python (pandas, NumPy, Matplotlib, seaborn), Jupyter Notebook
Analysis type	Exploratory and descriptive (Phase 1 of an ongoing study)
Prepared by	[Analyst name]
Date	October 2026

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1. Introduction

1.1 Background

Artificial intelligence has moved from a specialist research field to a core capability in finance, healthcare, retail, energy, telecommunications, gaming and many other industries. As organisations adopt AI, demand has grown for roles such as Machine Learning Engineer, Data Scientist, NLP Engineer and AI Engineer. Professionals, employers, educators and policy makers need evidence on how these roles are rewarded, where openings exist, and what working conditions look like.

This report documents an exploratory analysis of a global AI jobs dataset containing 90,000 job records described by 35 variables. The variables cover compensation, experience, education, industry, company characteristics, working conditions, hiring difficulty, automation risk and wellbeing indicators. All analysis, code and interpretation were carried out in Python.

1.2 Objectives

- Understand the structure, quality and coverage of the global AI jobs dataset.
- Organise the 35 variables into meaningful thematic segments and profile their distributions.
- Examine how salary and bonus vary by experience level and education, and over time.
- Identify the industries and countries with the largest volume of job openings and the hardest-to-fill roles.
- Measure the linear relationship between salary, bonus and company rating.
- Translate findings into practical recommendations and a roadmap for deeper analysis.

1.3 Importance of the Project

The project matters for several reasons. First, AI compensation is a high-stakes topic for job seekers deciding which skills and career paths to pursue, and for employers who must price roles competitively. Second, the dataset spans multiple countries, industries and seven years, which allows market-level questions to be examined from a global rather than a single-country perspective. Third, a clean, documented exploratory analysis is the necessary foundation for every later technique: modelling, forecasting and dashboarding all depend on knowing the data's quality, distributions and limitations. Finally, the work demonstrates a reproducible analytical workflow (load, audit, segment, visualise, interpret) that can be reused on similar labour-market data.

1.4 Scope and Continuation of the Analysis

> **This analysis will continue**
> This report covers the first phase of work on the dataset. With 35 columns and 90,000 rows, the dataset contains far more information than can be exhausted in a single analysis. Many further studies are possible, including regression analysis, machine learning, clustering, hypothesis

> testing, time-series analysis and interactive dashboards. These will be carried out in subsequent phases, and this report should be read as the foundation on which they will be built.

1.5 Tools and Environment

Tool	Purpose in this project
Python 3 / Jupyter Notebook	Interactive development environment for code, output and documentation
pandas	Loading the CSV, inspection, grouping, aggregation and type conversion
NumPy	Numerical support for pandas operations
Matplotlib	Base plotting: pie chart, bar charts, dual-axis line chart, figure export
seaborn	Statistical plots: histograms with KDE, box plots, count plots, heatmap, pair plot

2. Story of the Data

2.1 What the Dataset Describes

The file global_ai_jobs.csv records AI-related job postings and the surrounding labour-market context. Each row is one record with a unique identifier running from 1 to 90,000. Records span the years 2020 to 2026 (mean year 2023.0), so the data captures both historical and forward-dated observations. Every row combines information about the job (role, specialisation, experience, education), the employer (industry, size, funding, AI adoption), the location (country, cost of living, tax rate) and the experience of working there (hours, vacation, satisfaction, job security).

2.2 Structure

Property	Value
Rows (records)	90,000
Columns (variables)	35
Categorical (object) columns	8: country, job_role, ai_specialization, experience_level, education_required, industry, company_size, work_mode
Integer columns	18
Float columns	9
Missing values	0 in every column
Duplicate rows	0
Memory footprint	About 24 MB

2.3 Coverage of the Categorical Variables

- **Job roles (8):** NLP Engineer, Software Engineer AI, Research Scientist, Machine Learning Engineer, AI Engineer, Computer Vision Engineer, Data Scientist and Data Analyst.
- **AI specialisations (8):** Computer Vision, LLM, Reinforcement Learning, Analytics, Generative AI, MLOps, Forecasting and NLP.
- **Experience levels (4):** Entry, Mid, Senior and Lead.
- **Education required (5):** Bootcamp, Diploma, Bachelor, Master and PhD.
- **Company size (5):** Startup, Small, Medium, Large and Enterprise.
- **Work mode (3):** Onsite, Remote and Hybrid.
- **Industries:** Automotive, Retail, Healthcare, Tech, Consulting, Telecom, Energy, Gaming, Education and Finance.
- **Countries:** at least twelve appear in the sampled rows, including USA, UK, Canada, Germany, France, Netherlands, India, Japan, Singapore, Australia, UAE and Brazil.

2.4 The Story the Data Tells

Taken together, the dataset tells a story of a global, multi-industry AI labour market in which pay rises sharply with seniority, yet many other features, such as openings per company size, hiring difficulty per industry and salary per education level, look strikingly similar across groups. This near-uniformity is itself an important finding and is discussed in Sections 5 and 6, because it shapes what the data can and cannot be used to conclude.

3. Data Splitting and Preprocessing

3.1 Importing and Loading

The analysis begins by importing pandas, NumPy, Matplotlib and seaborn, then reading global_ai_jobs.csv into a DataFrame with pd.read_csv. The DataFrame is inspected immediately with df, df.info(), df.columns, df.describe(), df.head(15) and df.tail(15) to confirm that the load succeeded and the schema matches expectations.

3.2 Logical Splitting of the Data into Segments

With 35 columns, a single plot of everything would be unreadable. The data was therefore split by column into five thematic segments held in a Python dictionary. This split is a logical (analytical) split rather than a train-test split: every segment still contains all 90,000 rows, and each segment groups the variables that answer one family of questions.

Segment	Variables
Compensation & Finance	salary_usd, bonus_usd, salary_percentile, tax_rate_percent, cost_of_living_index
Wellbeing & Work-Life	work_life_balance_score, employee_satisfaction, job_security_score, layoff_risk, weekly_hours, vacation_days
Career Growth	career_growth_score, promotion_speed, experience_years, skill_demand_score, automation_risk
Hiring & Market	interview_rounds, hiring_difficulty_score, job_openings, offer_acceptance_rate, company_rating
Company & Economy	company_funding_billion, ai_adoption_score, ai_maturity_years, economic_index

A reusable function, plot_segment(), then profiles each segment automatically. For each column it selects the most suitable chart: a horizontal bar chart of the top 10 categories for text columns, a count plot for numeric columns with 15 or fewer distinct values, and a histogram with a KDE curve for continuous numeric columns. Writing this logic once and looping over the dictionary keeps the notebook compact, consistent and easy to extend.

3.3 Preprocessing Checks

Step	Method	Result
Missing values	df.isna().sum()	None in any of the 35 columns
Duplicates	df.duplicated().sum()	0 duplicate rows
Data types	df.info()	8 object, 18 int64, 9 float64
Range check	df.describe()	All values fall in plausible ranges (e.g. salary 28,000 to 300,622)
Type conversion	pd.to_datetime(df['weekly_hours'])	Applied in the notebook; see the note below

> **Note on the weekly_hours conversion**
> weekly_hours holds numeric working hours (36 to 55), so it is naturally a float. Passing numeric values to pd.to_datetime interprets them as nanosecond offsets from 1970 rather than hours, so the converted column no longer represents hours. It is recorded here for transparency. For the next phase, weekly_hours should be kept as float64 (reload the CSV or cast back) so it can be used in regression and machine-learning work.

3.4 Train-Test Splitting

No train-test split was needed in this phase because the work is descriptive. When regression and machine-learning models are developed in later phases, the data will be split into training and test sets (for example 80/20 with a fixed random seed, and stratification where a categorical target is used), after which categorical variables will be encoded and numeric variables scaled using parameters learned from the training set only, to avoid data leakage.

3.5 Output Management

Each chart is exported with plt.savefig() so that figures can be reused in this report, in Excel or PowerPoint, and in a Power BI dashboard.

4. Pre-Analysis

Pre-analysis is the audit stage: it establishes whether the data is complete, consistent and suitable for analysis before any conclusions are drawn.

4.1 Data Quality Assessment

- **Completeness:** 90,000 non-null values in each of the 35 columns; no imputation was required.
- **Uniqueness:** no duplicated rows; the id column is a clean sequence from 1 to 90,000.
- **Validity:** minimum and maximum values are within sensible bounds. Scores lie within 0 to 100, vacation days within 10 to 30, company rating within 3.2 to 4.8 and experience years within 0 to 19.
- **Consistency:** column types match their meaning, with the exception of the weekly_hours conversion described in Section 3.3.

4.2 Descriptive Statistics

Table 1 summarises the numeric variables for which the notebook output is complete (the remaining columns are available in the notebook and can be regenerated with df.describe()).

Variable	Mean	Std. dev.	Min	Median	Max
experience_years	7.03	5.89	0	6	19
salary_usd	96,546	43,935	28,000	87,544	300,622
bonus_usd	13,028	7,887	1,404	11,279	57,681
interview_rounds	4.5	1.7	2	4	7
year	2,023	2	2,020	2,023	2,026
weekly_hours	45.48	5.48	36	45.5	55
company_rating	4	0.46	3.2	4	4.8
job_openings	17.52	7.85	1	17	50
hiring_difficulty_score	55.03	17.9	0	55.07	100
vacation_days	19.99	6.07	10	20	30
skill_demand_score	50.46	28.85	1	51	100
automation_risk	50.36	28.85	1	50	100
job_security_score	75.56	11.32	29	77	99
career_growth_score	57.2	12.9	25	57	99
work_life_balance_score	69.15	13.21	25	69	98
promotion_speed	38.44	18.43	12	37	98
salary_percentile	50.54	28.89	1	51	100
cost_of_living_index	1.5	0.58	0.5	1.51	2.5
employee_satisfaction	72.73	8.12	42	73	99

Table 1: Summary statistics of numeric variables (from df.describe()).

4.3 Early Observations

- **Salary is right-skewed.** The mean salary (96,546 USD) is about 9,000 USD above the median (87,544 USD), and the maximum (300,622 USD) is more than three times the mean. A small number of very high salaries pull the average upwards.
- **Bonus is also right-skewed.** Mean 13,028 USD against a median of 11,279 USD. The average bonus equals roughly 13.5% of the average salary.
- **Many scores look uniformly spread.** skill_demand_score, automation_risk and salary_percentile each have a mean near 50 and quartiles near 25, 50 and 75 across a 1 to 100 range, which is the signature of a near-uniform distribution. tax_rate_percent and cost_of_living_index appear flat in their histograms too.
- **Wellbeing indicators are comparatively high.** Mean job security is 75.6, work-life balance 69.1 and employee satisfaction 72.7, against career growth at 57.2 and promotion speed at 38.4.

Because so many variables appear uniform and independent, the dataset may be synthetically generated or simulated. This should be verified with the data provider, and it should be borne in mind when interpreting every result in this report.

5. In-Analysis

This section describes the analytical techniques applied and the results each one produced. The accompanying charts are in Section 7.

5.1 Univariate Profiling by Segment

Each of the five segments was profiled with histograms, count plots and bar charts (Figures 1 to 5). In the Compensation segment, salary and bonus show clear right-skewed distributions, with most salaries between about 50,000 and 130,000 USD and a long tail towards 300,000 USD. By contrast, salary_percentile, tax_rate_percent and cost_of_living_index are almost flat across their ranges, meaning each value band is about equally represented.

5.2 Categorical Composition

- **Work mode:** the three modes are almost evenly split: Onsite 33.6%, Remote 33.3% and Hybrid 33.1% (Figure 6).
- **Job roles:** each of the eight roles has roughly 11,000 to 11,400 records. NLP Engineer has the most and Data Analyst the fewest (Figure 7).
- **AI specialisations:** Computer Vision and LLM lead at about 11,500 records each, with NLP lowest at about 11,000 (Figure 8).

5.3 Salary by Experience Level

Box plots of salary by experience level (Figure 9) show the clearest pattern in the data. Approximate median salaries read from the chart are: Entry about 63,000 USD, Mid about 80,000 USD, Senior about 108,000 USD and Lead about 147,000 USD. A Lead professional therefore earns roughly 2.3 times the median entry-level salary. Spread also widens with seniority, and the Lead group has the highest outliers, reaching about 300,000 USD.

5.4 Salary and Job Security by Education

Box plots by education level (Figures 10 and 16) show that median salary is almost identical for Bootcamp, Diploma, Bachelor, Master and PhD requirements, at about 87,000 to 88,000 USD, with similar quartile ranges (about 65,000 to 124,000 USD). Median job security is 77 for every level. In this dataset, the education level required does not visibly separate salary or job security.

5.5 Job Openings by Company Size, Industry and Country

- **Company size:** the distribution of job openings is virtually identical across Startup, Small, Medium, Large and Enterprise, with a median of 17 and an interquartile range of 12 to 23 (Figure 12).
- **Industry:** Gaming has the most total openings (about 161,000) and Automotive the fewest of the ten (about 156,000). The gap between first and last is only around 3% (Figure 13).

- **Country:** Singapore leads the top-ten countries (about 133,500 openings), followed closely by Australia, Canada and the UK; Germany is last of the ten at about 130,000 (Figure 14).

5.6 Hiring Difficulty by Industry

Average hiring difficulty is about 55 in every industry. Education is marginally the highest and Energy marginally the lowest, but the differences are too small to be meaningful (Figure 15).

5.7 Time Trend of Salary and Bonus

The dual-axis line chart (Figure 11) shows average salary and bonus by year from 2020 to 2026. Average salary moves between about 96,230 USD (2022) and 96,890 USD (2020), a total spread of under 1%. Average bonus ranges from about 12,960 USD (2024) to 13,085 USD (2026), also under 1%. Both axes are zoomed in tightly, which makes the movements look dramatic on the chart, but in practice both measures are essentially flat over the period.

5.8 Correlation Analysis

A correlation matrix was calculated for salary_usd, bonus_usd and company_rating (Figure 17), supported by a pair plot (Figure 18). Salary and bonus have a strong positive correlation of 0.75, meaning higher-paid roles tend to carry larger bonuses. Company rating has essentially zero linear correlation with either salary (-0.0004) or bonus (0.0017).

6. Post-Analysis and Insights

6.1 Key Insights

1. **Seniority is the dominant visible driver of pay.** Median salary rises from roughly 63,000 USD at Entry level to roughly 147,000 USD at Lead level, the strongest and most interpretable pattern in the analysis.
2. **Education level shows no salary premium here.** Median salaries are almost the same for all five education levels, so credentials alone do not explain pay in this dataset.
3. **Bonus follows salary.** The 0.75 correlation and a bonus of about 13.5% of average salary suggest bonuses are structured as a proportion of base pay.
4. **The market looks balanced across roles, specialisations, work modes and industries.** No single role, specialisation or work mode dominates; differences in openings between top and bottom industries and countries are around 3%.
5. **Pay has been stable from 2020 to 2026.** Year-to-year changes in average salary and bonus are below 1%.
6. **Company rating is unrelated to pay.** Better-rated companies do not pay more or less in this data.
7. **Remote work is as common as onsite work.** The three work modes are each about one third of the records.

6.2 Interpretation Cautions

- Many variables are uniformly or evenly distributed, and several group comparisons show almost no difference. This is unusual for real-world labour data and suggests the dataset may be simulated. Findings should be described as patterns in this dataset, not as confirmed global labour-market facts, until the data source is verified.
- Correlation was assessed for only three variables. Relationships involving the other 30+ columns remain unexplored.
- Salary is recorded in USD without an explicit adjustment for purchasing power. Cost of living index is present and should be considered when comparing countries.
- Values in this report read from charts (medians by experience level, openings totals) are approximate. Exact values can be reproduced with groupby().median() and groupby().sum().

6.3 Summary of Findings

Question	Finding
Does experience affect salary?	Yes, strongly: about 63k (Entry) to about 147k (Lead) median
Does education affect salary?	No visible effect: medians about 87k to 88k for every level
Does bonus move with salary?	Yes: correlation 0.75
Does company rating relate to pay?	No: correlation close to 0

Question	Finding
Has pay changed over time?	Almost not at all: under 1% movement 2020 to 2026
Which industry has the most openings?	Gaming (about 161,000), only about 3% above Automotive
Which country has the most openings?	Singapore, among the top ten shown
Which work mode is most common?	Onsite 33.6%, essentially tied with Remote and Hybrid

7. Data Visualizations and Charts

All charts were produced with Matplotlib and seaborn and use a consistent navy palette. Each figure is followed by a short interpretation.

7.1 Segment Profiles

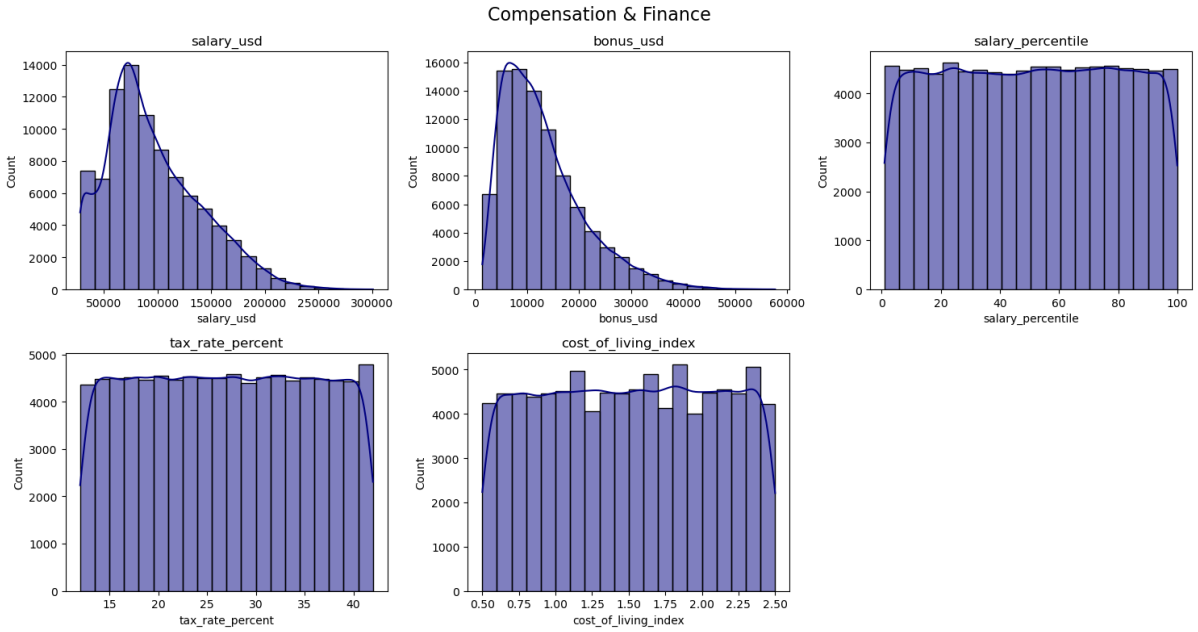


Figure 1: *Compensation & Finance segment: distributions of salary, bonus, salary percentile, tax rate and cost of living.*

Interpretation: Salary and bonus are right-skewed with long upper tails; salary percentile, tax rate and cost of living index are close to uniform.

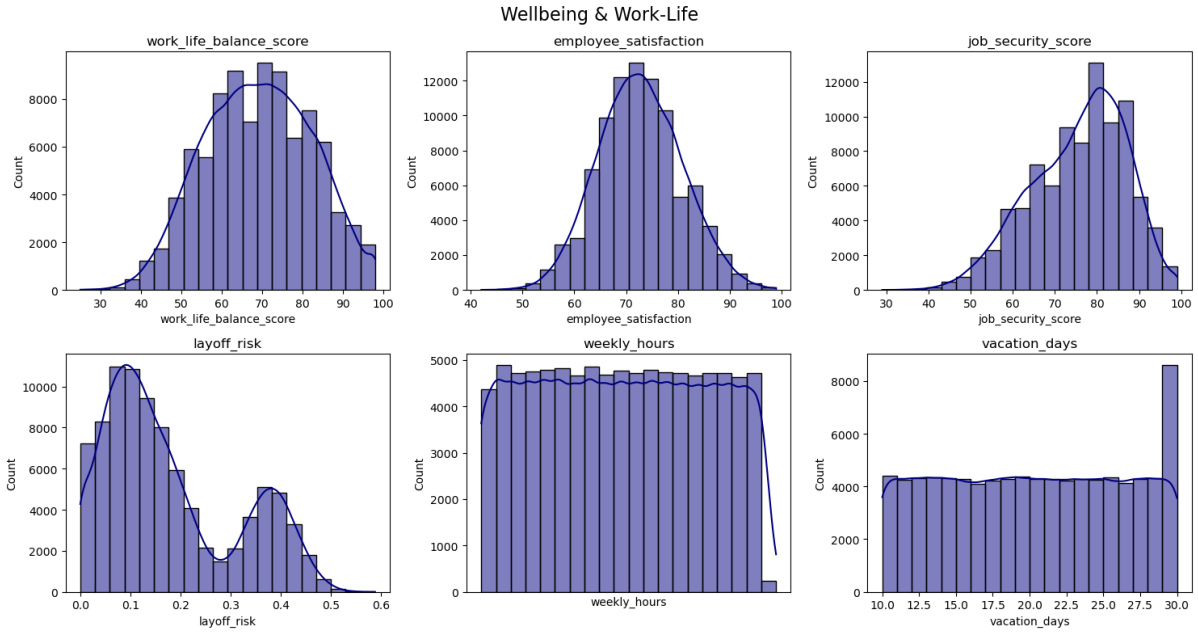


Figure 2: *Wellbeing & Work-Life segment: work-life balance, satisfaction, job security, layoff risk, weekly hours and vacation days.*

Interpretation: Distributions of the wellbeing indicators across all 90,000 records. Note that weekly_hours was converted to datetime in the notebook (Section 3.3).

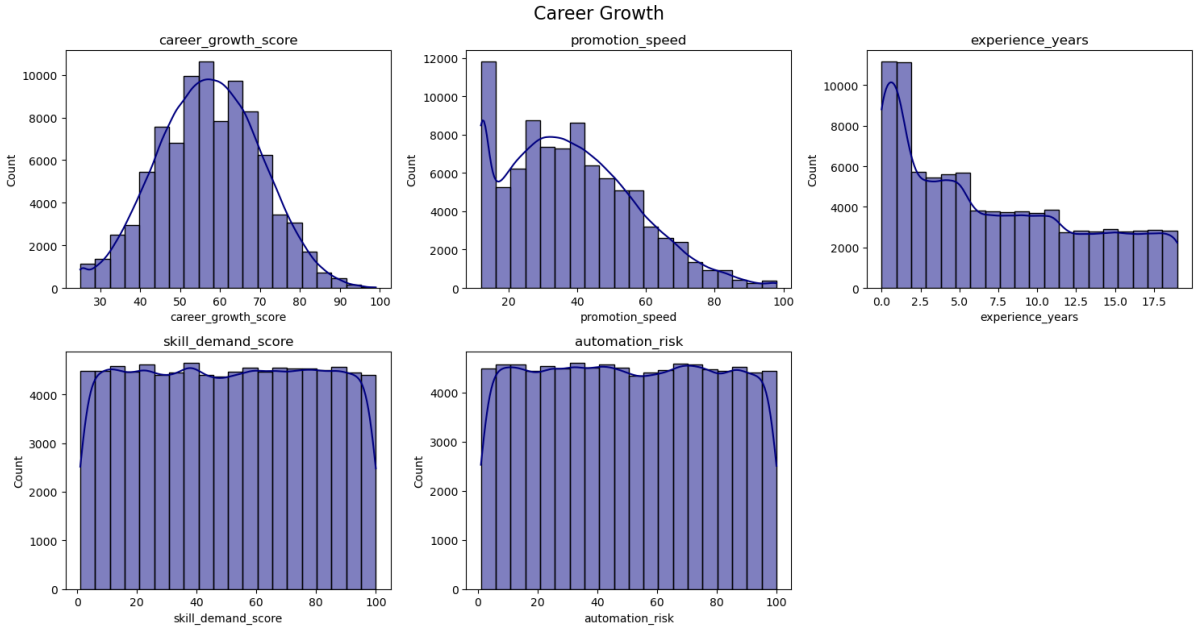


Figure 3: *Career Growth segment: career growth score, promotion speed, experience years, skill demand and automation risk.*

Interpretation: Profiles of career-related indicators; skill demand and automation risk are spread broadly across 1 to 100.

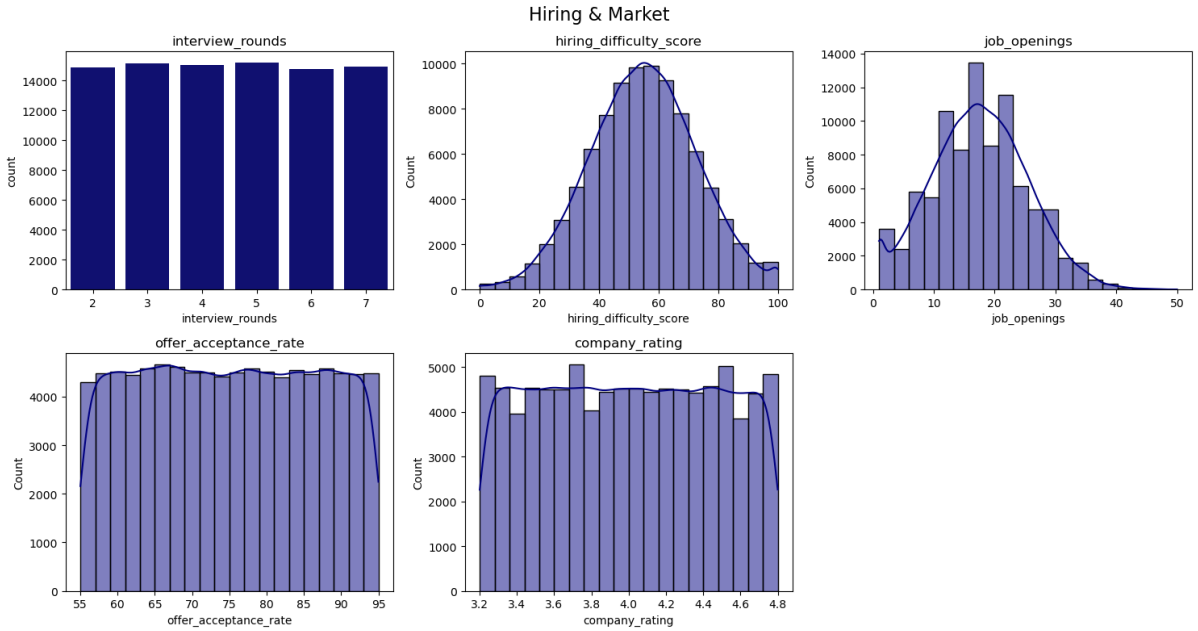


Figure 4: *Hiring & Market segment: interview rounds, hiring difficulty, job openings, offer acceptance rate and company rating.*

Interpretation: Shows the typical hiring process (2 to 7 interview rounds) and the spread of openings and hiring difficulty.

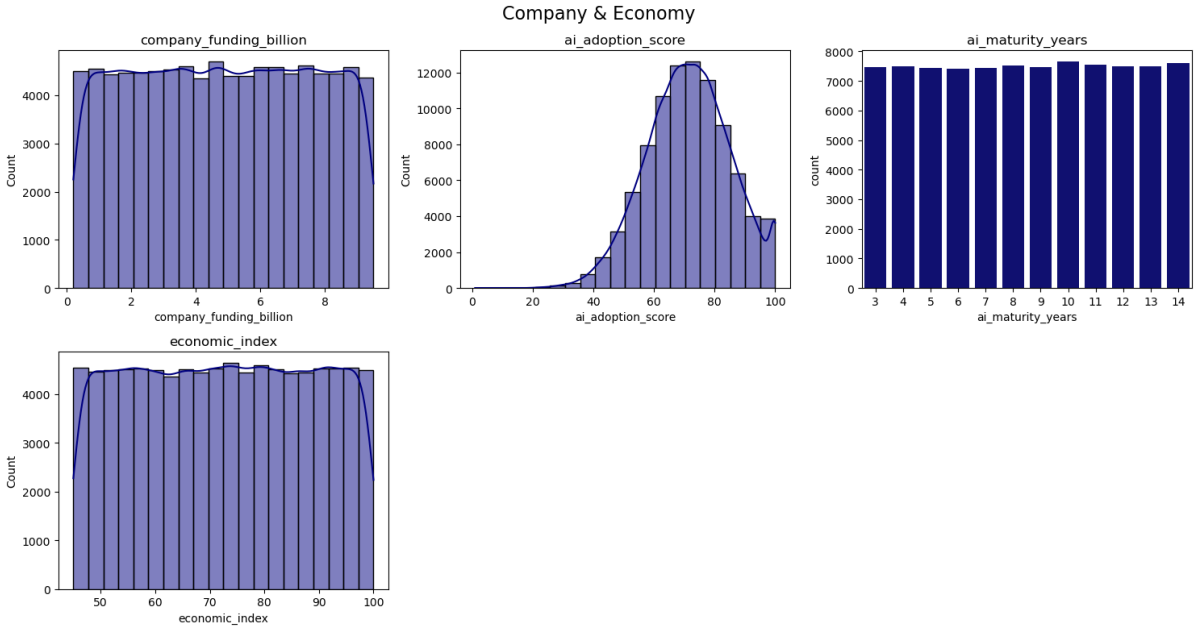


Figure 5: *Company & Economy segment: funding, AI adoption, AI maturity years and economic index.*

Interpretation: Distributions of the organisation-level and economic context variables.

7.2 Composition of the Market

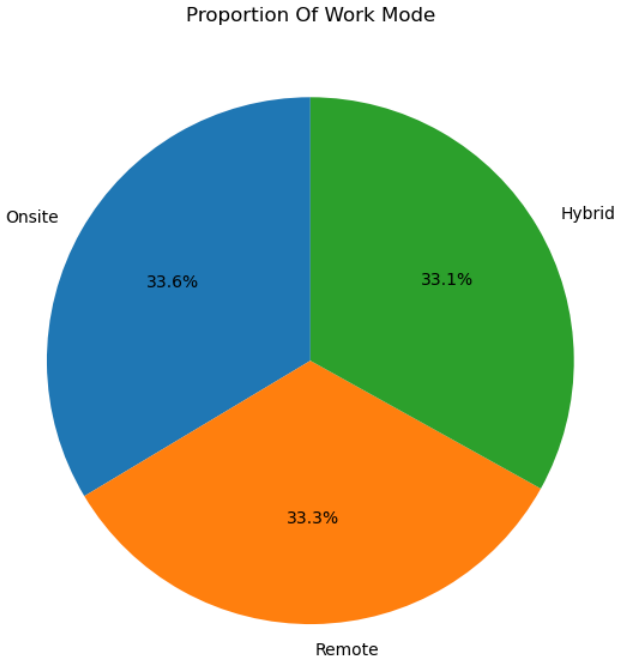


Figure 6: *Proportion of work mode.*

Interpretation: Onsite (33.6%), Remote (33.3%) and Hybrid (33.1%) are almost evenly split.

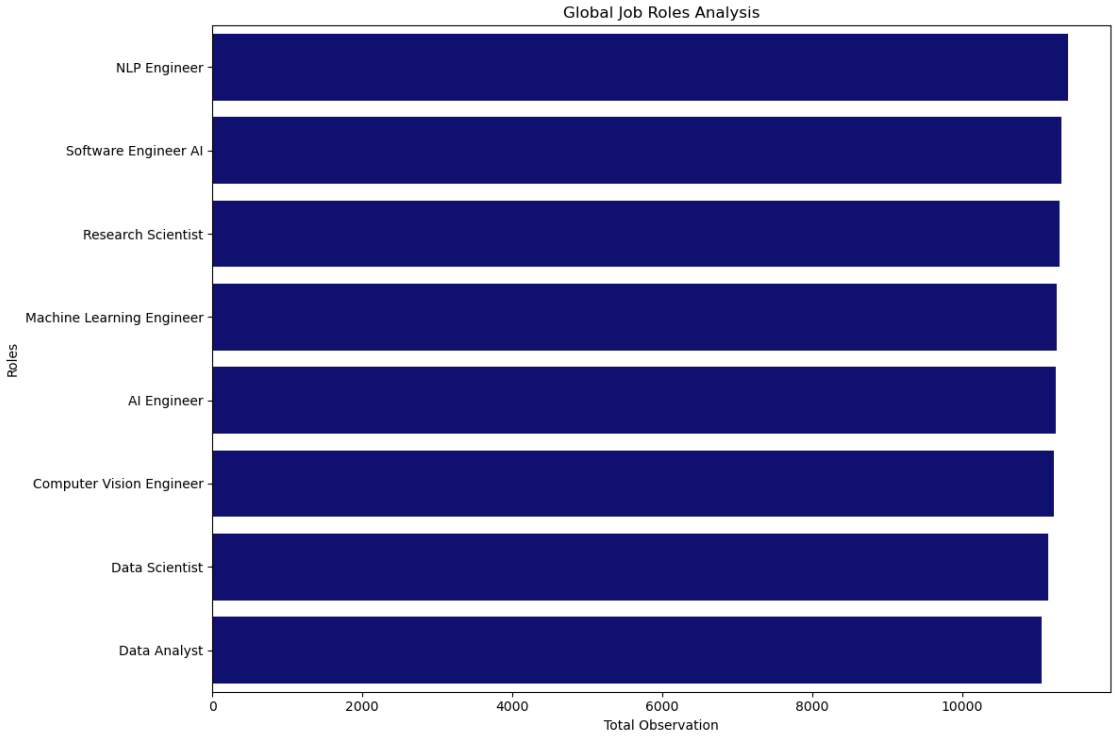


Figure 7: *Global job roles by number of observations.*

Interpretation: All eight roles are close in size (about 11,000 to 11,400 records); NLP Engineer is highest.

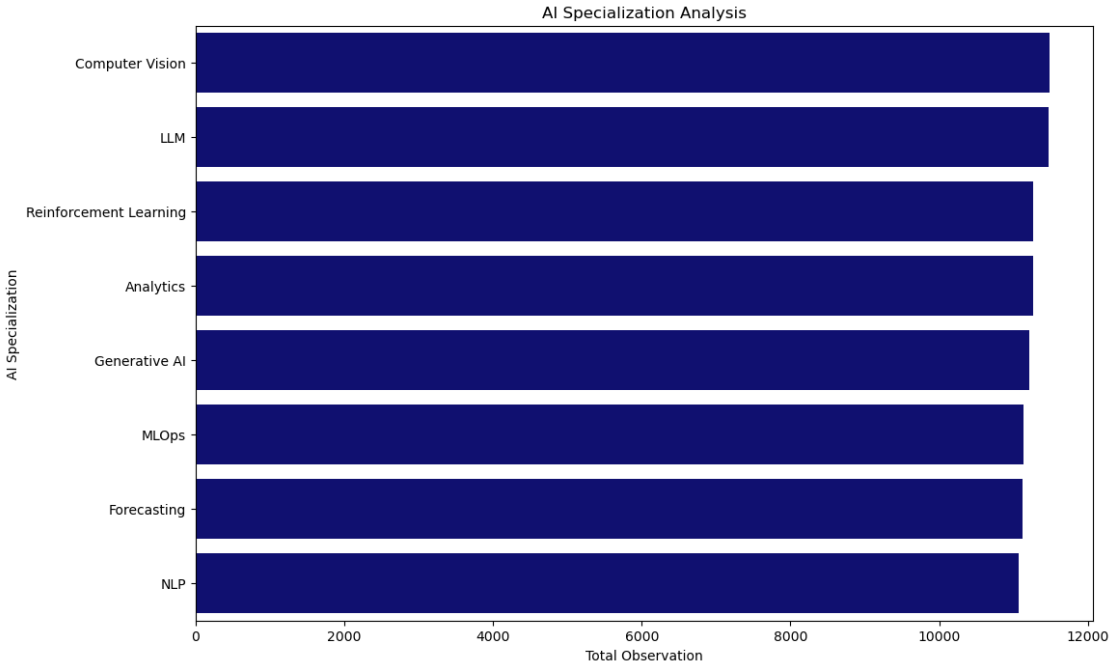


Figure 8: *AI specialisation by number of observations.*

Interpretation: Computer Vision and LLM are slightly most frequent; NLP is least frequent, but differences are small.

7.3 Compensation

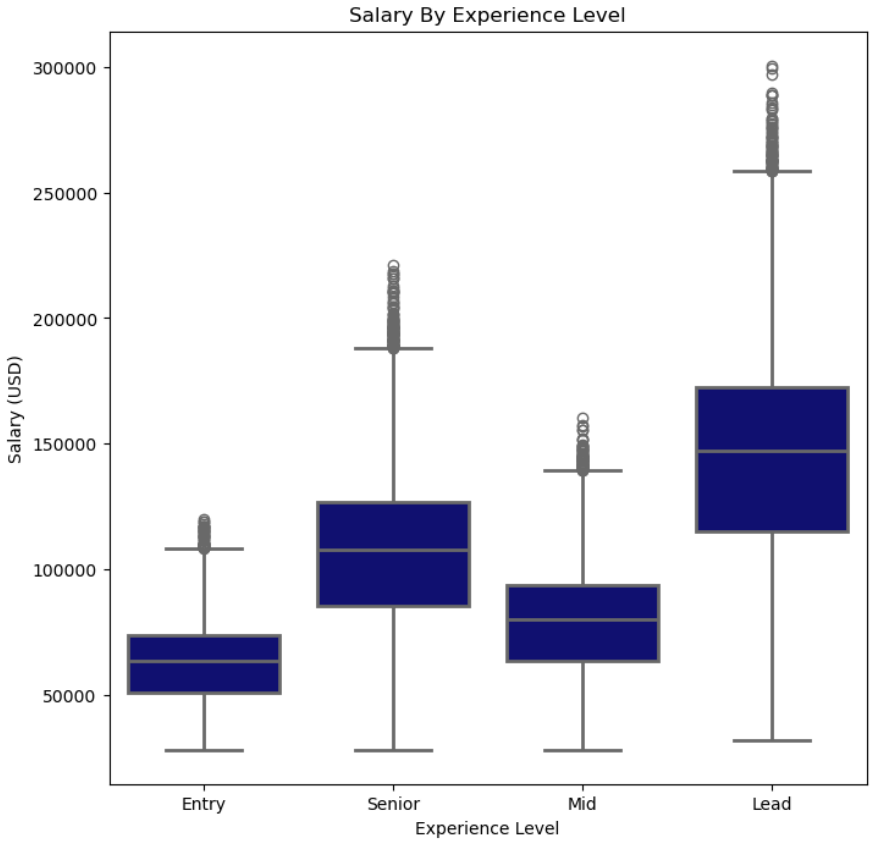


Figure 9: *Salary by experience level.*

Interpretation: Salary rises with seniority and the spread widens at Lead level, which also holds the highest outliers. Categories appear in the order Entry, Senior, Mid, Lead; ordering them Entry, Mid, Senior, Lead would improve readability.

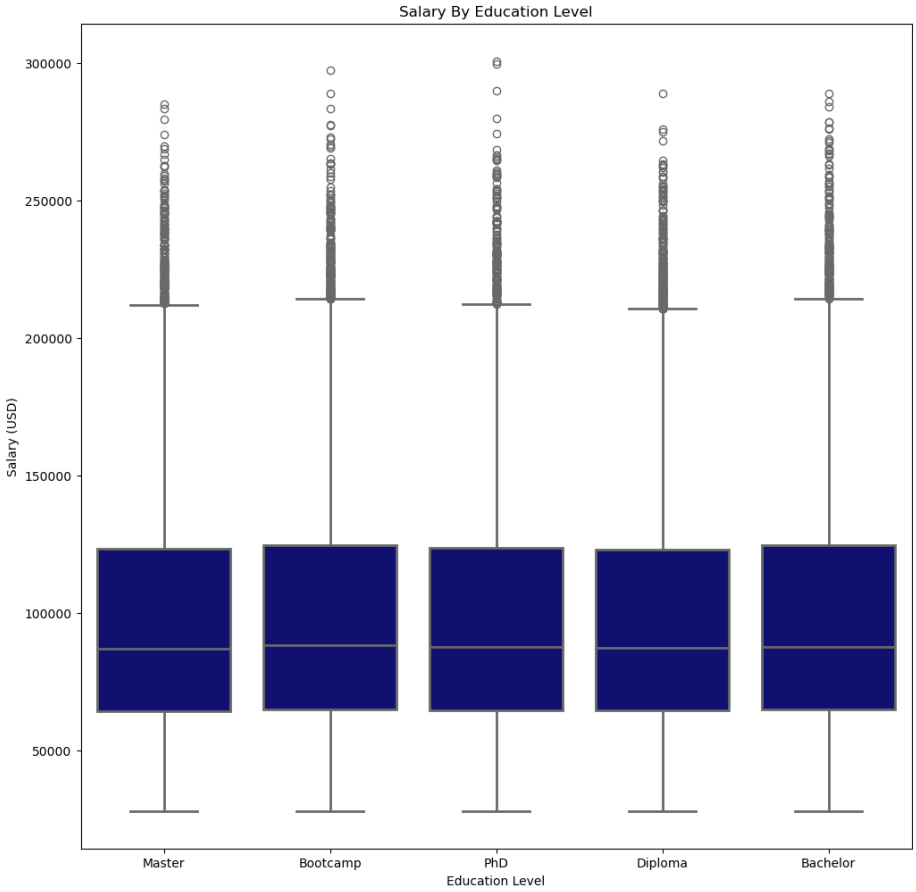


Figure 10: *Salary by education level.*

Interpretation: The five education levels have nearly identical medians, quartiles and outlier patterns.

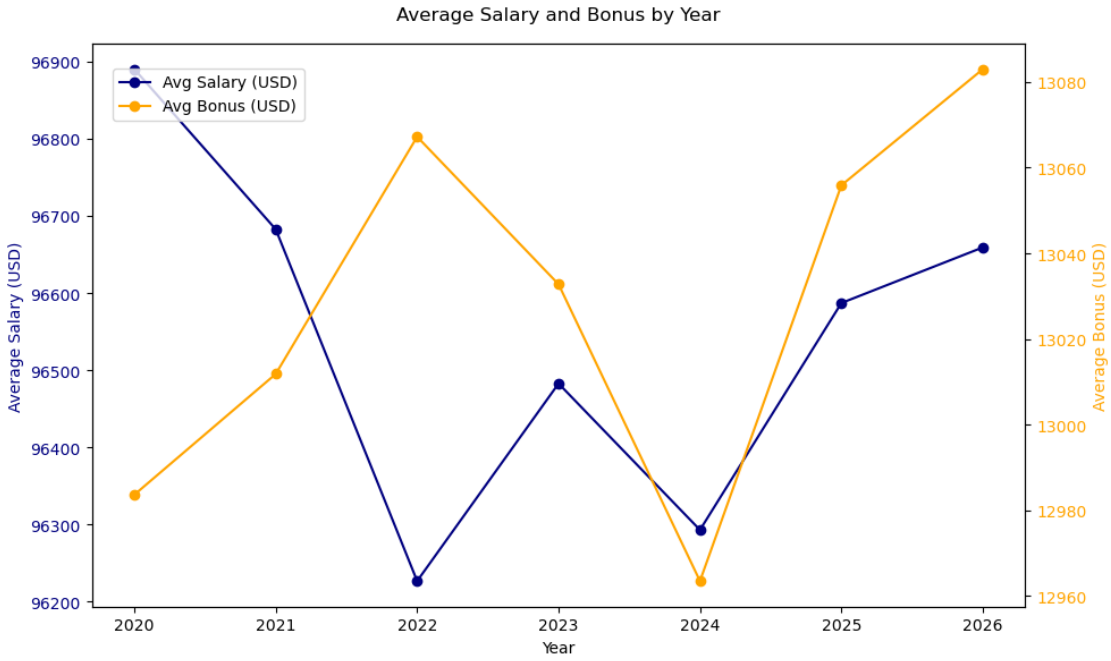


Figure 11: *Average salary and bonus by year, 2020 to 2026.*

Interpretation: Both measures change by less than 1% across the period; the narrow axis ranges exaggerate small movements.

7.4 Hiring and Market

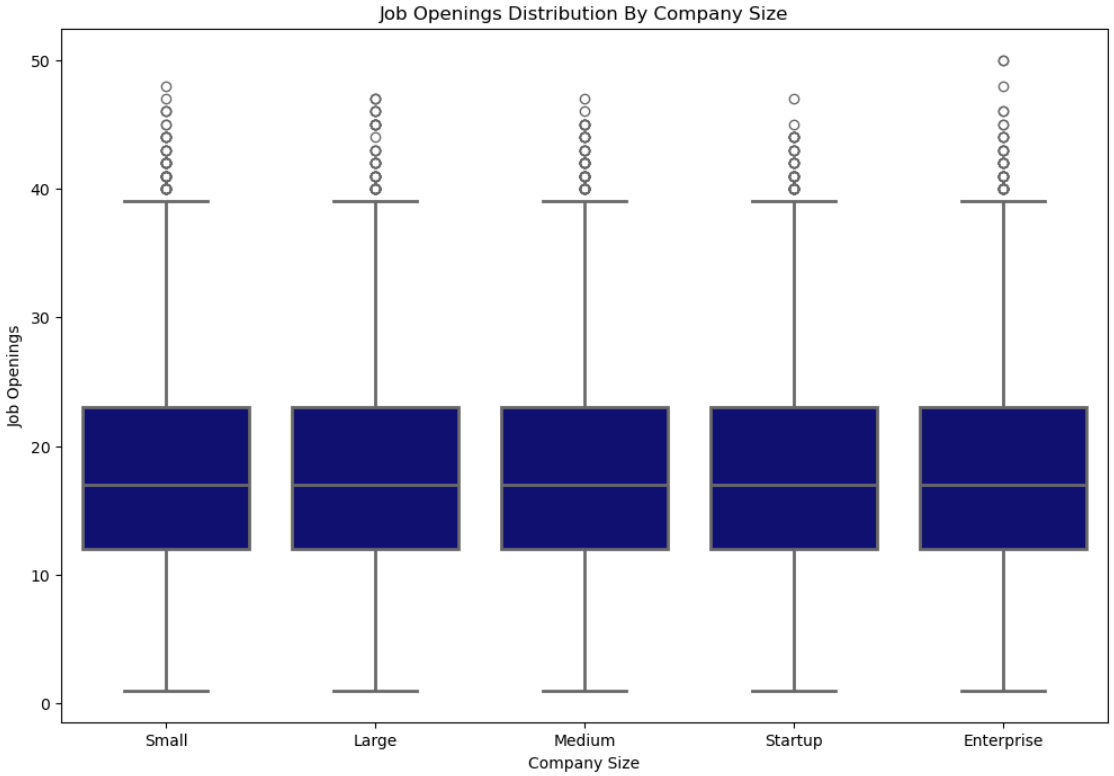


Figure 12: *Job openings distribution by company size.*

Interpretation: Identical medians (17) and interquartile ranges (12 to 23) in all five company-size groups.

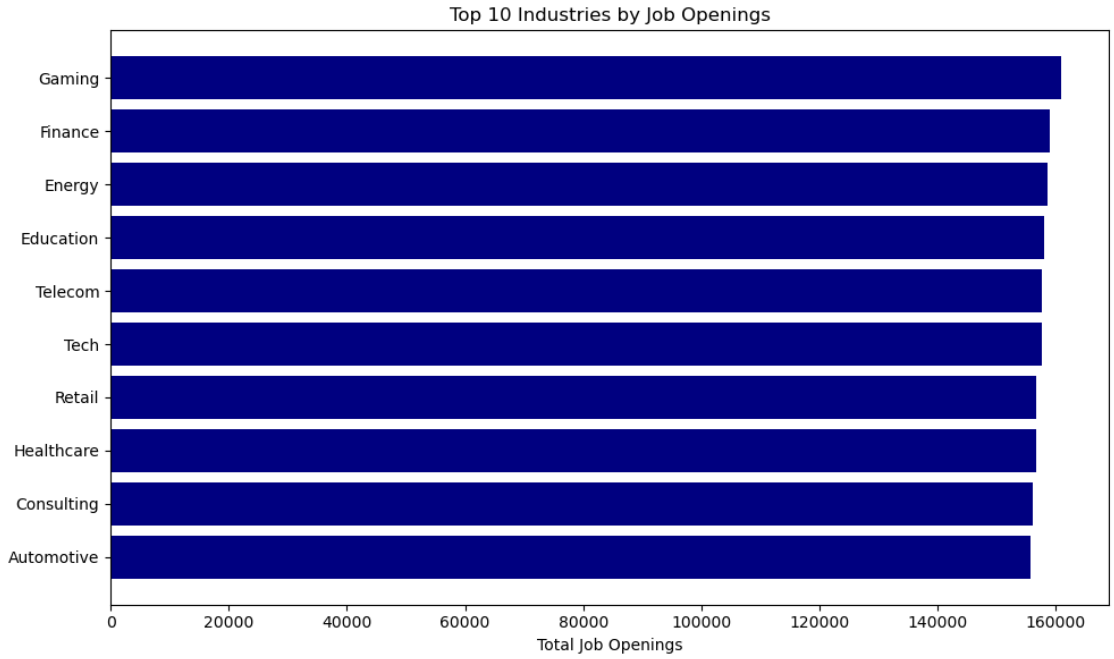


Figure 13: *Top 10 industries by total job openings.*

Interpretation: Gaming leads and Automotive is lowest, but the difference is only about 3%.

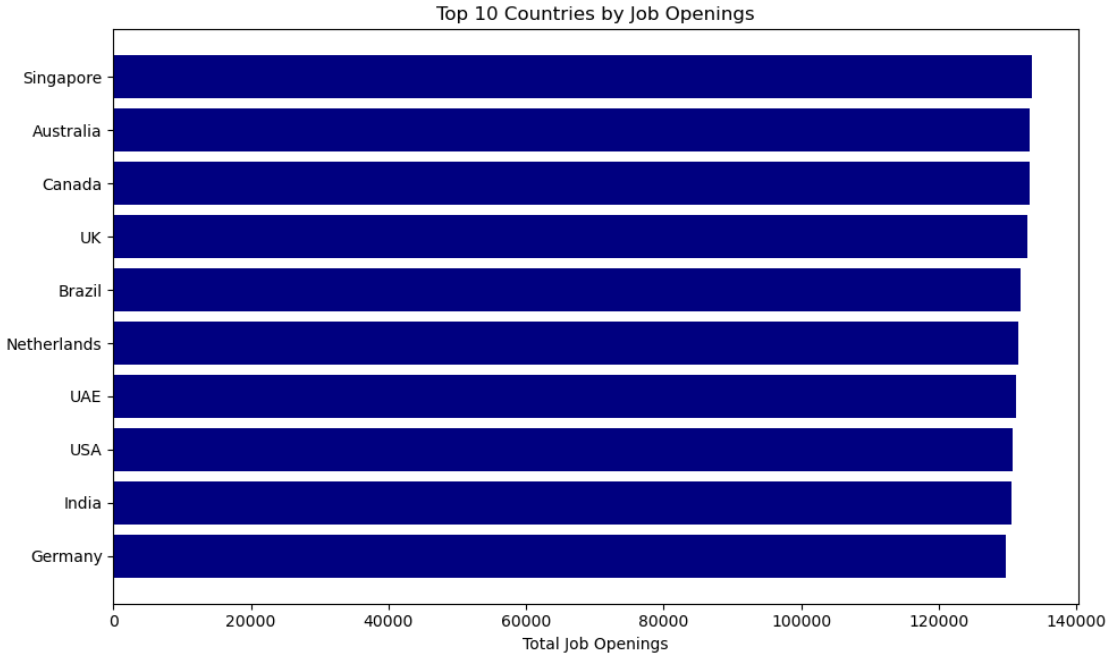


Figure 14: *Top 10 countries by total job openings.*

Interpretation: Singapore, Australia, Canada and the UK are at the top, in a tight range.

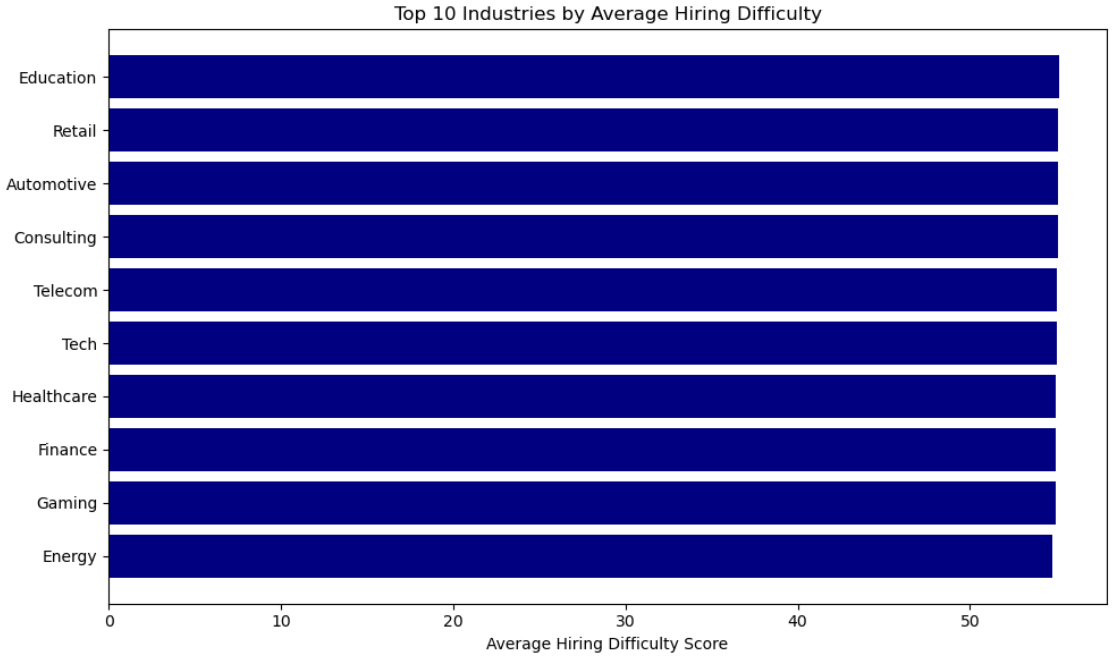


Figure 15: *Top 10 industries by average hiring difficulty score.*

Interpretation: All industries average about 55, so no industry is clearly harder to hire in.

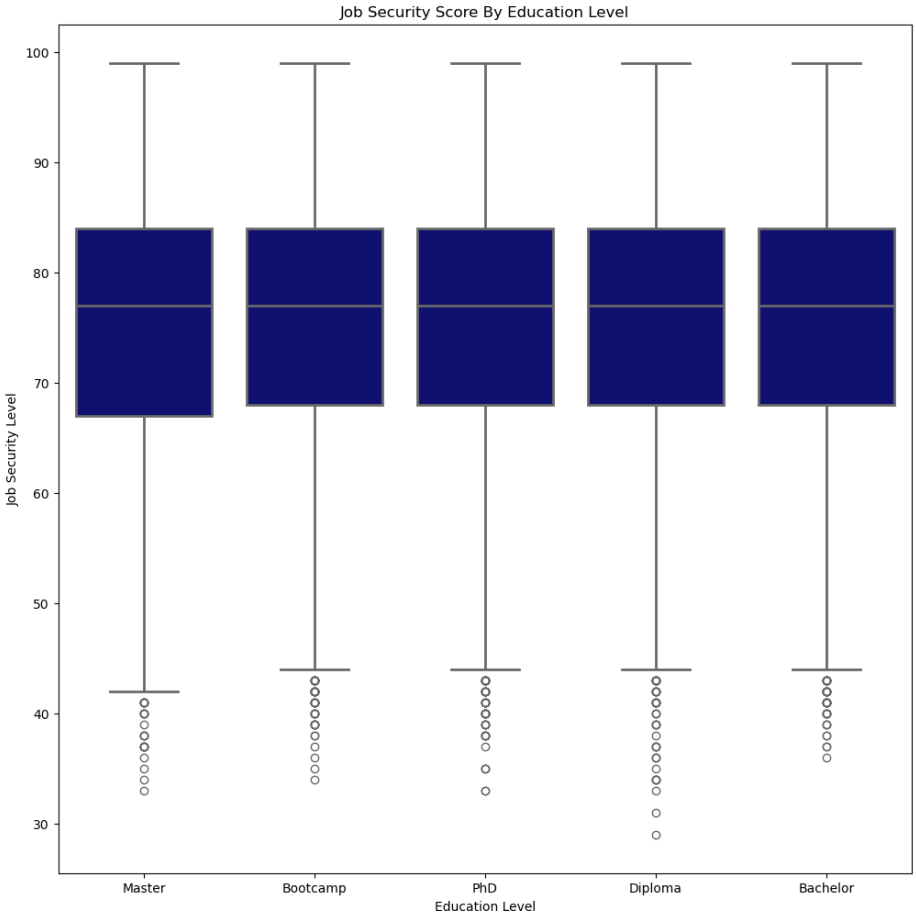


Figure 16: *Job security score by education level.*

Interpretation: Median job security is 77 for every education level.

7.5 Relationships

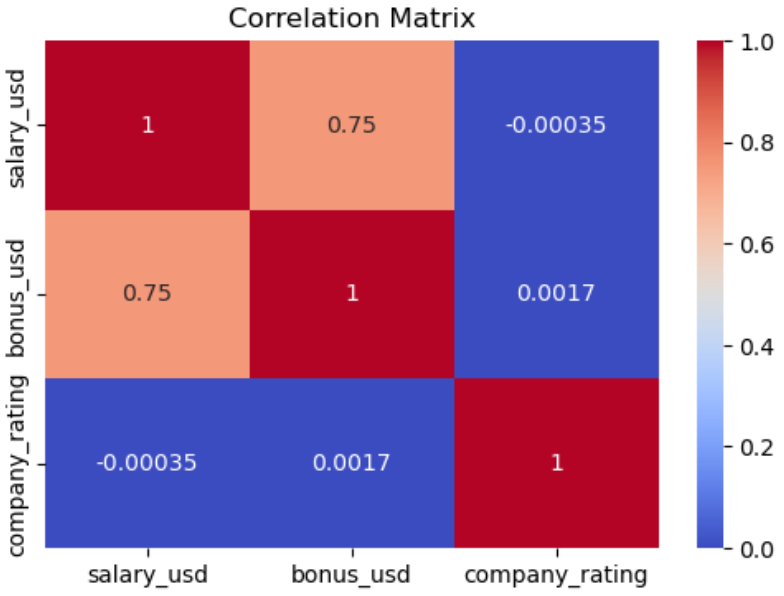


Figure 17: *Correlation matrix of salary, bonus and company rating.*

Interpretation: Salary and bonus correlate at 0.75; company rating is uncorrelated with both.

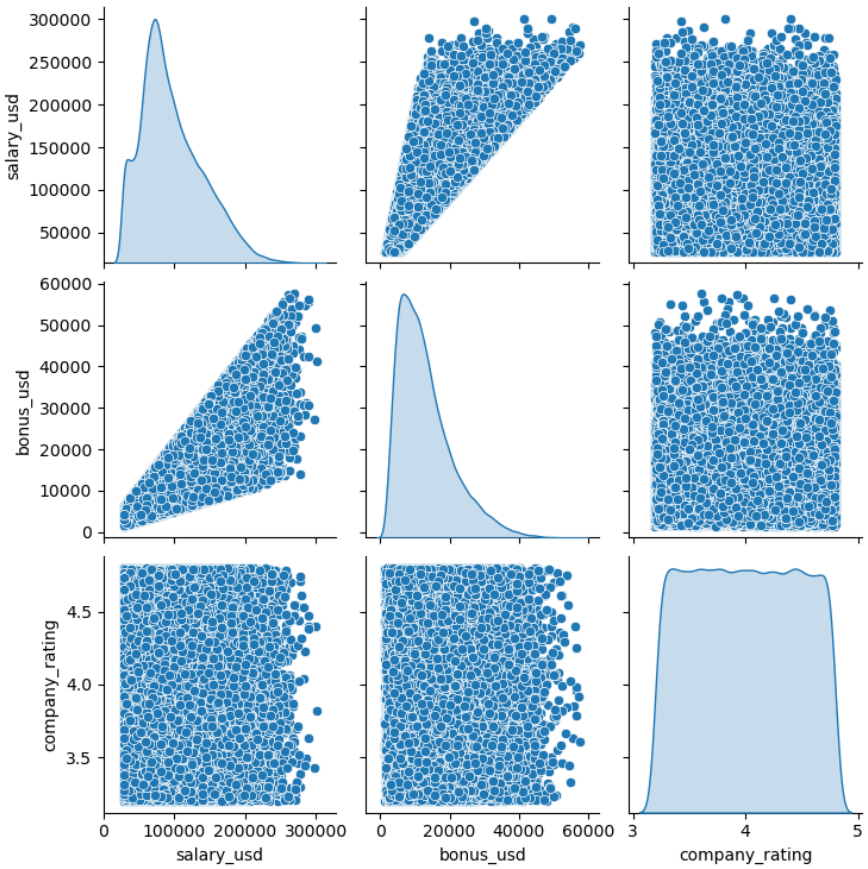


Figure 18: *Pair plot of salary, bonus and company rating.*

Interpretation: The scatter plots confirm a positive salary-bonus relationship and no pattern involving company rating.

8. Recommendations and Observations

8.1 Recommendations Based on the Findings

For job seekers and early-career professionals

- Plan for progression: moving from Entry to Mid, Senior and Lead is associated with the largest increases in pay in this dataset.
- Do not assume a higher qualification alone raises pay; the data shows no salary difference between education levels, so combine education with experience and demonstrable skills.
- With Onsite, Hybrid and Remote roles equally represented, work mode can be chosen on lifestyle fit rather than availability.

For employers and HR teams

- Benchmark compensation against experience level first, since that is the clearest pay differentiator.
- Consider structuring bonuses relative to base salary (about 13.5% on average), consistent with the observed salary-bonus relationship.
- Because hiring difficulty and openings are similar across industries, hiring strategy can be designed on a common baseline and refined by role.

For analysts and researchers (next steps)

- Verify the dataset's origin and generation method before drawing market-level conclusions.
- Extend the correlation analysis to all numeric variables and test relationships formally (ANOVA or Kruskal-Wallis for group differences, with effect sizes).
- Adjust salary for cost of living and tax rate to build a comparable real-income measure across countries.
- Build an interactive Power BI dashboard and SQL views so stakeholders can filter by country, role, industry and year.

8.2 Planned Future Analyses

The dataset supports many further studies, and this analysis will be continued. Planned directions include:

Study	Possible approach
Regression analysis	Predict salary_usd or bonus_usd from experience, role, specialisation, country, industry and company features (linear, regularised and tree-based regression)
Machine learning (classification)	Predict experience_level, work_mode or high versus low automation risk; evaluate with cross-validation and feature importance
Clustering	Group jobs by compensation, wellbeing and growth profile (K-Means, hierarchical clustering) to find job archetypes

Study	Possible approach
Hypothesis testing	Formally test differences in salary, satisfaction and job security by education, work mode and company size
Time-series analysis	Study trends in salary, openings and AI adoption from 2020 to 2026 by country and industry
Dashboarding	Power BI and Excel dashboards on top of cleaned data, with SQL as the data layer

8.3 Technical Observations on the Notebook

The following observations are recorded to keep the analysis reproducible and to guide the next phase.

- **weekly_hours conversion:** keep as float64 (see Section 3.3).
- **Saving figures:** plt.savefig() is called after plt.show(), and Matplotlib clears the figure once it is shown. This is why the notebook prints "<Figure size 640x480 with 0 Axes>" after the segment charts. Calling savefig before show, and using a unique file name with an extension (for example f"{name}.png") inside the segment loop, will save each chart correctly and prevent overwriting.
- **Category ordering:** use order=["Entry","Mid","Senior","Lead"] in the experience box plot.
- **Axis scaling on the yearly chart:** consider starting the axis at zero or annotating the percentage change to avoid overstating small movements.
- **File path:** the CSV is read from an absolute local path; a relative path or a configurable variable improves portability.

9. Conclusion

This report documented a complete exploratory analysis of the Global AI Jobs dataset using Python. The data is complete and clean, with 90,000 records, 35 variables, no missing values and no duplicates. After organising the variables into five segments, the analysis found that seniority is the dominant visible driver of salary, that bonus rises with salary (correlation 0.75), that education level and company rating show no visible relationship with pay, and that pay has been essentially flat between 2020 and 2026. Openings, hiring difficulty, work modes, roles and specialisations are all spread very evenly across categories.

The very even distributions and near-zero correlations are themselves a key finding: they suggest that the dataset may be simulated and that its findings should be treated as patterns within the dataset until its source is confirmed.

This is the first phase of an ongoing body of work. The dataset's breadth of 35 columns and 90,000 rows means it holds far more information than one analysis can exhaust. The next phases will apply regression analysis, machine learning, clustering, hypothesis testing, time-series analysis and dashboarding to extract deeper insight. The groundwork in this report (data audit, segmentation, reusable plotting functions and documented limitations) is designed to make those studies faster and more reliable.

10. References and Appendices

10.1 References

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Kluyver, T., Ragan-Kelley, B., Perez, F., et al. (2016). Jupyter Notebooks: a publishing format for reproducible computational workflows. In Positioning and Power in Academic Publishing: Players, Agents and Agendas (pp. 87-90). IOS Press.

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Appendix A: Data Dictionary

Data types are taken from df.info(). Descriptions are interpreted from the column names and observed value ranges and should be confirmed against the dataset's official documentation.

#	Column	Type	Interpreted description
1	id	int64	Unique record identifier (1 to 90,000)
2	country	object	Country of the job posting
3	job_role	object	AI-related job title (8 roles)
4	ai_specialization	object	AI specialisation area (8 areas)
5	experience_level	object	Entry, Mid, Senior or Lead
6	experience_years	int64	Years of experience (0 to 19)
7	salary_usd	int64	Annual salary in US dollars
8	bonus_usd	int64	Annual bonus in US dollars
9	education_required	object	Education level required (Bootcamp, Diploma, Bachelor, Master, PhD)
10	industry	object	Industry of the employer
11	company_size	object	Startup, Small, Medium, Large or Enterprise
12	interview_rounds	int64	Number of interview rounds (2 to 7)
13	year	int64	Year of the record (2020 to 2026)
14	work_mode	object	Onsite, Remote or Hybrid
15	weekly_hours	float64	Weekly working hours (36 to 55)
16	company_rating	float64	Company rating (3.2 to 4.8)
17	job_openings	int64	Number of openings (1 to 50)

#	Column	Type	Interpreted description
18	hiring_difficulty_score	float64	Difficulty of filling the role (0 to 100)
19	layoff_risk	float64	Estimated layoff risk
20	ai_adoption_score	int64	Degree of AI adoption in the organisation
21	company_funding_billion	float64	Company funding in billions of USD
22	economic_index	float64	Economic condition indicator
23	ai_maturity_years	int64	Years of AI maturity of the organisation
24	offer_acceptance_rate	float64	Share of job offers accepted
25	tax_rate_percent	float64	Applicable tax rate (%)
26	vacation_days	int64	Annual vacation days (10 to 30)
27	skill_demand_score	int64	Demand for the role's skills (1 to 100)
28	automation_risk	int64	Risk of the role being automated (1 to 100)
29	job_security_score	int64	Job security (29 to 99)
30	career_growth_score	int64	Career growth opportunity (25 to 99)
31	work_life_balance_score	int64	Work-life balance (25 to 98)
32	promotion_speed	int64	Speed of promotion (12 to 98)
33	salary_percentile	int64	Salary percentile (1 to 100)
34	cost_of_living_index	float64	Cost of living index (0.5 to 2.5)
35	employee_satisfaction	int64	Employee satisfaction (42 to 99)

Appendix B: Summary Statistics

Statistics from df.describe() for the numeric variables with complete notebook output (see Table 1 in Section 4.2).

Variable	Mean	Std. dev.	Min	Median	Max
experience_years	7.03	5.89	0	6	19
salary_usd	96,546	43,935	28,000	87,544	300,622
bonus_usd	13,028	7,887	1,404	11,279	57,681
interview_rounds	4.5	1.7	2	4	7
year	2,023	2	2,020	2,023	2,026
weekly_hours	45.48	5.48	36	45.5	55
company_rating	4	0.46	3.2	4	4.8
job_openings	17.52	7.85	1	17	50
hiring_difficulty_score	55.03	17.9	0	55.07	100
vacation_days	19.99	6.07	10	20	30
skill_demand_score	50.46	28.85	1	51	100
automation_risk	50.36	28.85	1	50	100
job_security_score	75.56	11.32	29	77	99
career_growth_score	57.2	12.9	25	57	99
work_life_balance_score	69.15	13.21	25	69	98
promotion_speed	38.44	18.43	12	37	98
salary_percentile	50.54	28.89	1	51	100
cost_of_living_index	1.5	0.58	0.5	1.51	2.5
employee_satisfaction	72.73	8.12	42	73	99

Appendix C: Full Python Code

The complete code from the analysis notebook, in the order in which it was run.

Cell 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Cell 2

```
df = pd.read_csv(r"C:\Users\dorca\Downloads\global_ai_jobs.csv")
```

Cell 3

```
df
```

Cell 4

```
df.info()
```

Cell 5

```
df.columns
```

Cell 6

```
df.describe()
```

Cell 7

```
df.isna().sum()
```

Cell 8

```
df.duplicated().sum()
```

Cell 9

```
df.head(15)
```

Cell 10

```
df.tail(15)
```

Cell 11

```
df['weekly_hours'] = pd.to_datetime(df['weekly_hours'])
```

Cell 12

```
df.info()
```

Cell 13

```
segments = {
    'Compensation & Finance': ['salary_usd', 'bonus_usd', 'salary_percentile',
                               'tax_rate_percent', 'cost_of_living_index'],
    'Wellbeing & Work-Life': ['work_life_balance_score', 'employee_satisfaction',
                              'job_security_score', 'layoff_risk',
                              'weekly_hours', 'vacation_days'],
    'Career Growth': ['career_growth_score', 'promotion_speed', 'experience_years',
                      'skill_demand_score', 'automation_risk'],
    'Hiring & Market': ['interview_rounds', 'hiring_difficulty_score', 'job_openings',
                        'offer_acceptance_rate', 'company_rating'],
    'Company & Economy': ['company_funding_billion', 'ai_adoption_score',
                          'ai_maturity_years', 'economic_index'],
}

def plot_segment(df, title, cols, ncols=3):
    nrows = -(-len(cols) // ncols)              # ceiling division
    fig, axes = plt.subplots(nrows, ncols, figsize=(5 * ncols, 4 * nrows))
    axes = axes.flatten()


    for ax, col in zip(axes, cols):
        if df[col].dtype == 'object':           # categorical: top 10 only
            top = df[col].value_counts().head(10)
            sns.barplot(x=top.values, y=top.index, ax=ax, color='navy')
        elif df[col].nunique() <= 15:           # discrete numeric: count each value
            sns.countplot(x=df[col], ax=ax, color='navy')
        else:                                   # continuous: histogram + curve
            sns.histplot(df[col], bins=20, kde=True, ax=ax, color='navy')
```

```
        ax.set_title(col)

    for ax in axes[len(cols):]:                # hide unused slots
        ax.axis('off')

    fig.suptitle(title, fontsize=16)
    plt.tight_layout()
    plt.show()

for name, cols in segments.items():
    plot_segment(df, name, cols)
    plt.savefig("segments charts")
```

Cell 14

```
work_mode_proportion = df['work_mode'].value_counts()
plt.figure(figsize=(6,6))

plt.pie(
    work_mode_proportion,
    labels = work_mode_proportion.index,
    autopct = '%1.1f%%',
    startangle = 90
)
plt.title('Proportion Of Work Mode')
plt.tight_layout()
plt.savefig("Porportion OF Work Mode")
plt.show()
```

Cell 15

```
Roles = df['job_role']
roles_counts = pd.Series(Roles).value_counts()
```

Cell 16

```
plt.figure(figsize=(12,8))
sns.barplot(x=roles_counts.values, y=roles_counts.index, color='navy')
plt.title('Global Job Roles Analysis')
plt.xlabel('Total Observation')
plt.ylabel('Roles')
plt.tight_layout()
plt.savefig("Global Job Roles chart")
plt.show()
```

Cell 17

```
Specialization = df['ai_specialization']
Specialization
specialization_counts = pd.Series(Specialization).value_counts()
```

Cell 18

```
plt.figure(figsize= (12,8))
sns.barplot(x=specialization_counts.values, y=specialization_counts.index, color='navy')
plt.title('AI Specialization Analysis')
plt.xlabel('Total Observation')
plt.ylabel('AI Specialization')
plt.savefig("AI Specialization chart")
plt.show()
```

Cell 19

```
plt.figure(figsize=(8,8))
sns.boxplot(x='experience_level', y='salary_usd', color='navy', linecolor='dimgray', linewidth= 2,
data=df)
plt.title('Salary By Experience Level')
plt.xlabel('Experience Level')
plt.ylabel('Salary (USD)')
plt.savefig("Salary By Experience Chart")
plt.show()
```

Cell 20

```
plt.figure(figsize=(12,8))
sns.boxplot(x='company_size', y='job_openings', color='navy', linecolor='dimgray', linewidth=
2,data=df)
plt.title('Job Openings Distribution By Company Size')
```

```
plt.xlabel('Company Size')
plt.ylabel('Job Openings')
plt.savefig("Job Distribution chart")
plt.show()
```

Cell 21

```
yearly = df.groupby('year')[['salary_usd', 'bonus_usd']].mean().sort_index()

fig, ax1 = plt.subplots(figsize=(10, 6))

ax1.plot(yearly.index, yearly['salary_usd'], marker='o', color='navy', label='Avg Salary (USD)')
ax1.set_xlabel('Year')
ax1.set_ylabel('Average Salary (USD)', color='navy')
ax1.tick_params(axis='y', labelcolor='navy')

ax2 = ax1.twinx()
ax2.plot(yearly.index, yearly['bonus_usd'], marker='o', color='orange', label='Avg Bonus (USD)')
ax2.set_ylabel('Average Bonus (USD)', color='orange')
ax2.tick_params(axis='y', labelcolor='orange')

fig.suptitle('Average Salary and Bonus by Year')
fig.legend(loc='upper left', bbox_to_anchor=(0.1, 0.9))
plt.tight_layout()
plt.savefig("Average Salary and Bonus by Year chart")
plt.show()
```

Cell 22

```
top_industries = df.groupby('industry')['job_openings'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_industries.index[::-1], top_industries.values[::-1], color='navy')
plt.title('Top 10 Industries by Job Openings')
plt.xlabel('Total Job Openings')
plt.tight_layout()
plt.savefig("Top 10 Industries by Job Openings Chart")
plt.show()
```

Cell 23

```
top_countries = df.groupby('country')['job_openings'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_countries.index[::-1], top_countries.values[::-1], color='navy')
plt.title('Top 10 Countries by Job Openings')
plt.xlabel('Total Job Openings')
plt.tight_layout()
plt.savefig("Countries by job openings chart")
plt.show()
```

Cell 24

```
top_difficulty =
df.groupby('industry')['hiring_difficulty_score'].mean().sort_values(ascending=False).head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_difficulty.index[::-1], top_difficulty.values[::-1], color='navy')
plt.title('Top 10 Industries by Average Hiring Difficulty')
plt.xlabel('Average Hiring Difficulty Score')
plt.tight_layout()
plt.savefig("10 Industries by average hiring difficulty chart")
plt.show()
```

Cell 25

```
plt.figure(figsize=(12,12))
sns.boxplot(x='education_required', y='salary_usd', color='navy', linecolor='dimgray', linewidth= 2,
data=df)
plt.title("Salary By Education Level")
plt.xlabel("Education Level")
plt.ylabel("Salary (USD)")
plt.savefig("Salary by education level chart")
plt.show()
```

Cell 26

```
plt.figure(figsize=(12,12))
sns.boxplot(x='education_required', y='job_security_score', color='navy', linecolor='dimgray',
linewidth= 2, data=df)
plt.title("Job Security Score By Education Level")
plt.xlabel("Education Level")
plt.ylabel("Job Security Level")
plt.savefig("job security score chart")
plt.show()
```

Cell 27

```
numeric_cols = ["salary_usd", "bonus_usd"]
if "company_rating" in df.columns:
    numeric_cols.append("company_rating")
```

Cell 28

```
corr = df[numeric_cols].corr()
plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.savefig("Correlation chart")
plt.show()
```

Cell 29

```
sns.pairplot(df[numeric_cols], kind="scatter", diag_kind="kde")
plt.savefig("pairplot chart for global aijobs")
plt.show()
```

Appendix D: SQL Equivalents of Key Aggregations

The main aggregations from the notebook, expressed as SQL for use if the dataset is loaded into a database (assumes a table named global_ai_jobs).

Top 10 industries by job openings

```
SELECT industry, SUM(job_openings) AS total_openings
FROM global_ai_jobs
GROUP BY industry
ORDER BY total_openings DESC
LIMIT 10;
```

Top 10 countries by job openings

```
SELECT country, SUM(job_openings) AS total_openings
FROM global_ai_jobs
GROUP BY country
ORDER BY total_openings DESC
LIMIT 10;
```

Average salary and bonus by year

```
SELECT year, AVG(salary_usd) AS avg_salary, AVG(bonus_usd) AS avg_bonus
FROM global_ai_jobs
GROUP BY year
ORDER BY year;
```

Average hiring difficulty by industry

```
SELECT industry, AVG(hiring_difficulty_score) AS avg_difficulty
FROM global_ai_jobs
GROUP BY industry
ORDER BY avg_difficulty DESC
LIMIT 10;
```

Average salary by experience level

```
SELECT experience_level, AVG(salary_usd) AS avg_salary
FROM global_ai_jobs
GROUP BY experience_level;
```